

GROUP WORK PROJECT # 1

MScFE 622: Stochastic Modeling

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Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an “X” above).

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Note: You may be required to provide proof of your outreach to non-contributing members upon request.

N/A - all members contributed.

Executive summary

This report presents the assumptions, results, interpretations, and figures for all project questions. SM Energy spot is \$232.90; the annual risk-free rate is 1.50%; one year is 250 trading days; and no dividend yield is assumed.

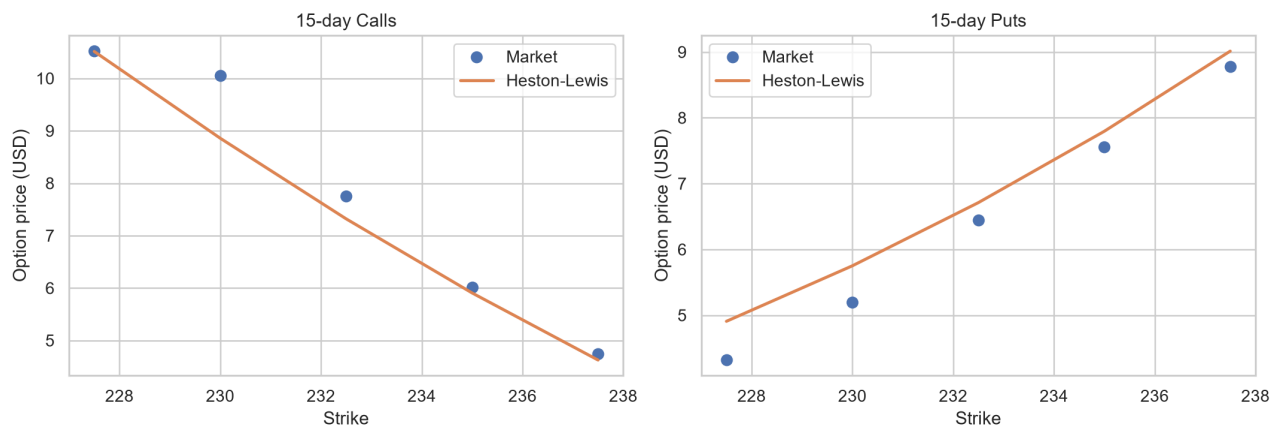
Step 1 - 15-day derivative

1(a) Heston calibration using Lewis (2001)

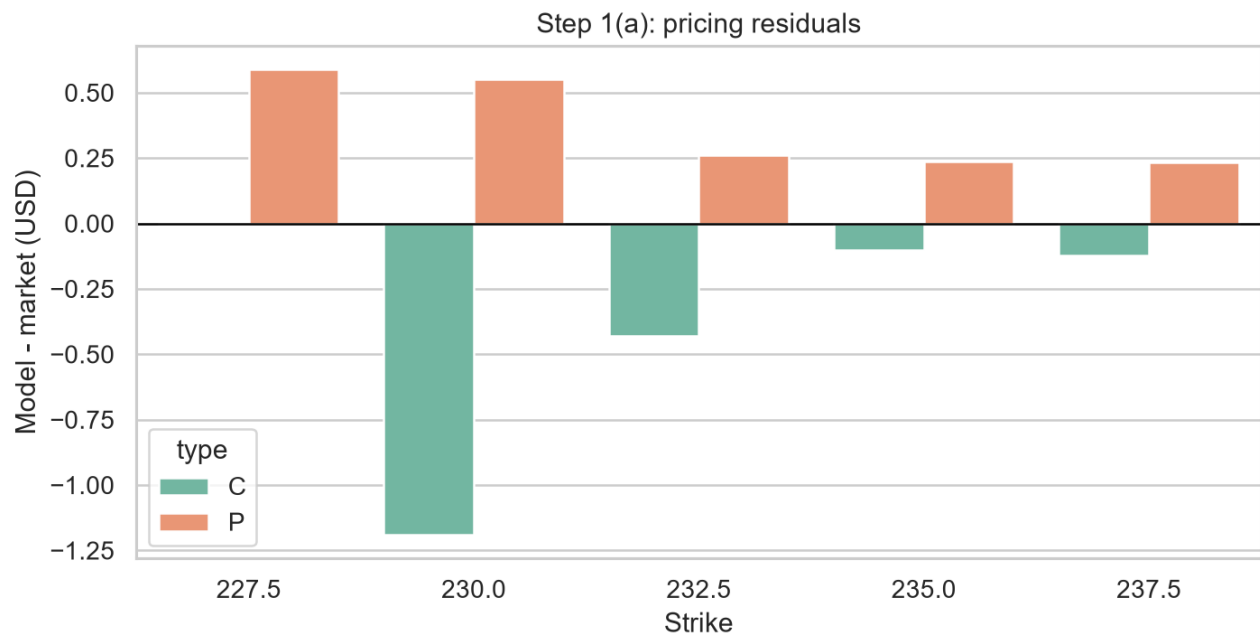
Heston represents equity returns with a randomly changing, mean-reverting variance process that can reproduce volatility clustering and implied-volatility skew. Lewis converts its characteristic function into a one-dimensional Fourier pricing integral.

The 15-day calls and puts were fitted jointly using the Heston specification and Lewis representation (Heston 327-343; Lewis), with unweighted USD price MSE as required by the project brief (WorldQuant University 1-2). The fitted parameters are $\kappa=2.2032$, $\theta=0.0010$, $\sigma=1.2733$, $\rho=-0.9444$, and initial variance=0.1084. MSE is 0.2458 and RMSE is \$0.4958.

The input quotes have a put-call parity RMSE of \$0.9068; an arbitrage-consistent fit cannot make all residuals zero. Short maturity also weakens identification of long-run variance and mean reversion.



15-day market and model prices

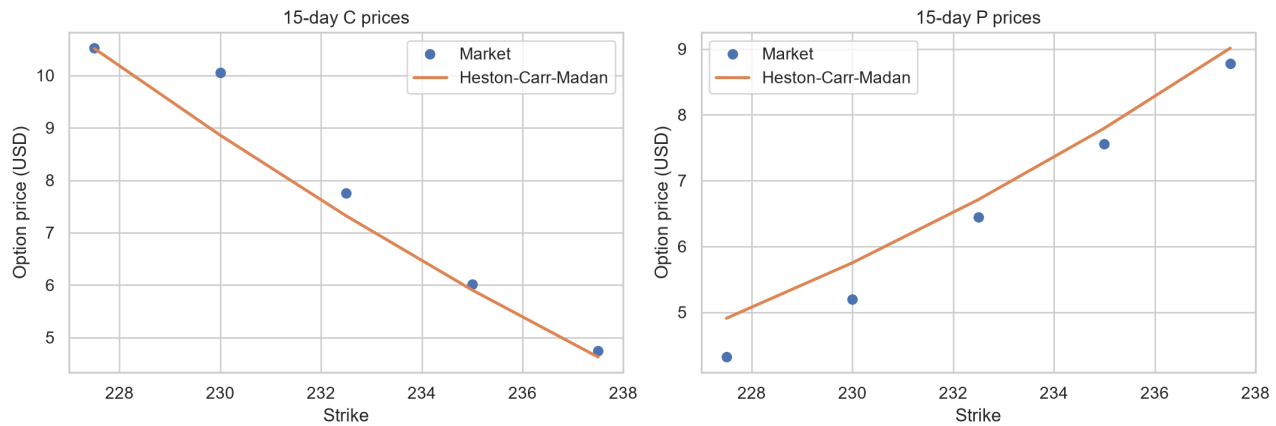


15-day residuals

1(b) Carr-Madan comparison

Carr-Madan damps the option-price function so it can be evaluated through an integrable Fourier transform, providing an independent pricing and calibration route for the same Heston dynamics.

The damped transform follows Carr and Madan (61-73). Its MSE is 0.2458, compared with 0.2458 for Lewis. Fitted prices and numerical stability are the meaningful comparison.

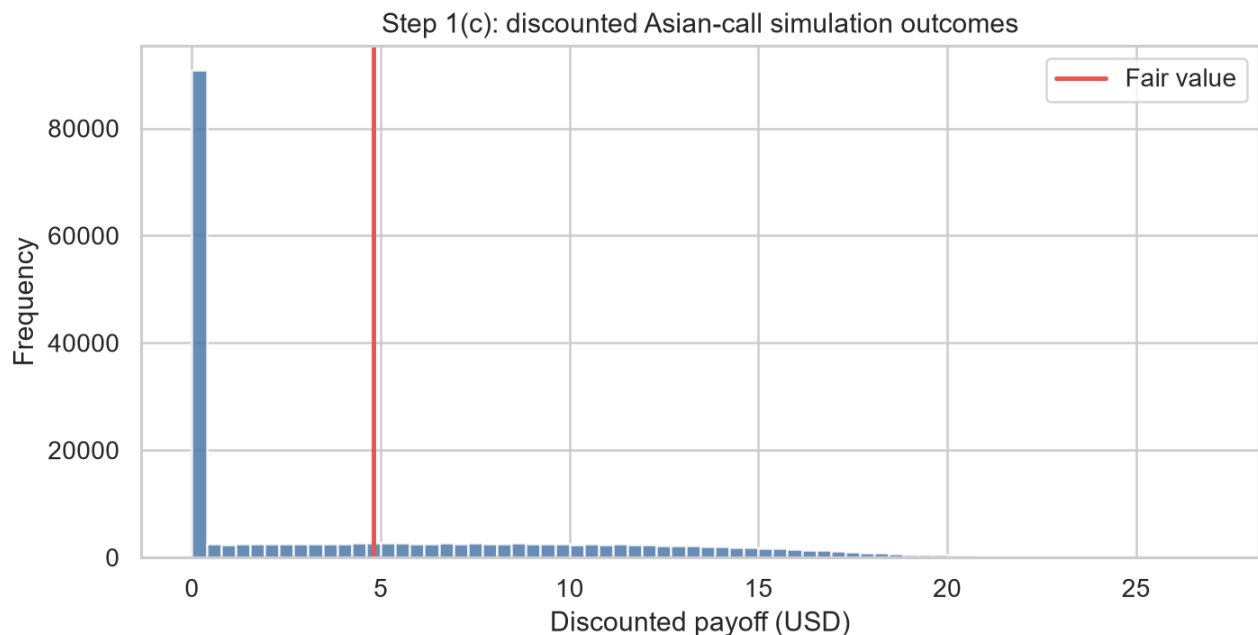


15-day Carr-Madan fit

1(c) 20-day ATM Asian call

The arithmetic-average Asian payoff reduces dependence on a single terminal stock price. Monte Carlo simulation is required because the payoff depends on the complete daily path.

Today's spot is included in the arithmetic average, following the project instructions (WorldQuant University 1-2). Across 200,000 risk-neutral scenarios, fair value is \$4.8177 (standard error \$0.0130), with a 95% estimator interval of \$4.7922 to \$4.8432. The 4% fee produces a client price of \$5.0104.



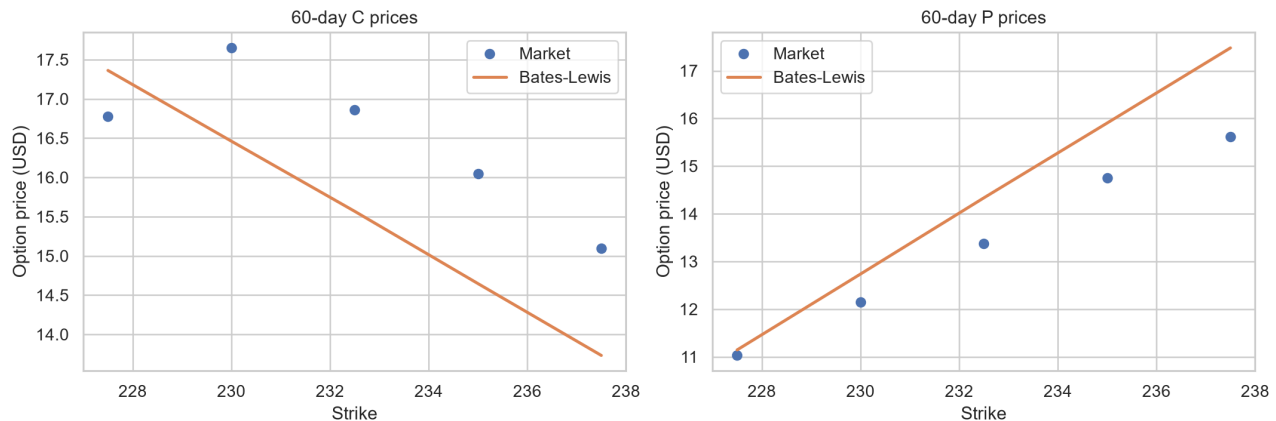
Asian-call outcome distribution

Step 2 - 60-day derivative

2(a) Bates calibration using Lewis

Bates extends Heston with random discontinuous price jumps, allowing the model to represent rare abrupt moves in addition to ordinary stochastic-volatility dynamics.

The 60-day jump-diffusion calibration follows Bates (69-107) and has MSE 1.3324 and RMSE \$1.1543.

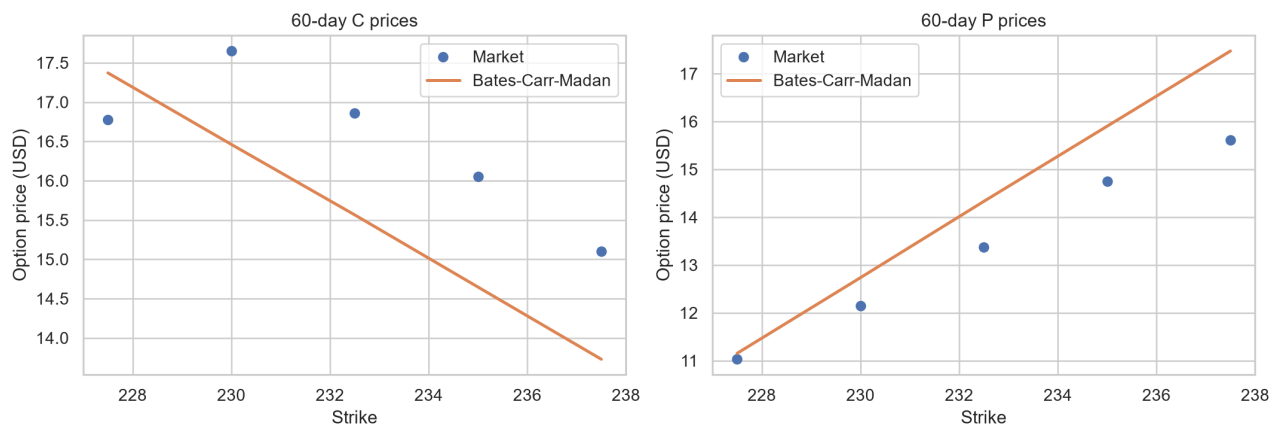


60-day market and model prices

2(b) Carr-Madan comparison

This specification retains the Bates jump and variance dynamics but uses the Carr-Madan transform as a numerical cross-check of Lewis pricing.

The matching Carr-Madan calibration follows Carr and Madan (61-73) and has MSE 1.3335. The comparison is based on fitted-price errors and numerical stability.



60-day Carr-Madan fit

2(c) 70-day 95%-moneyness put

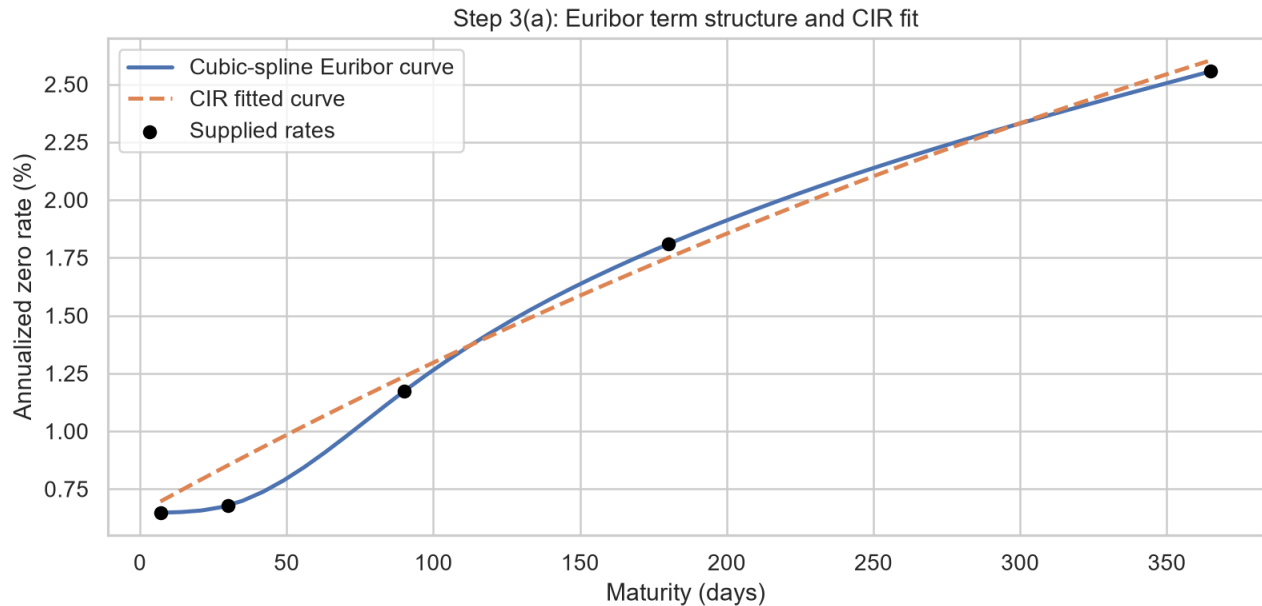
The project specifies a strike of 95% of spot (WorldQuant University 3), giving \$221.25. Fair value is \$8.8187 and the price after the 4% fee is \$9.1715.

Step 3 - Euribor rates

3(a) CIR calibration

CIR models a non-negative, mean-reverting short rate and connects its parameters to the Euribor curve through closed-form zero-coupon bond prices.

The supplied curve contains 0.648% (1 week), 0.679% (1 month), 1.173% (3 months), 1.809% (6 months), and 2.556% (12 months). Weekly interpolation follows the project brief (WorldQuant University 3-4), and the rate model follows Cox, Ingersoll, and Ross (385-407). The fitted values are $a=0.7525$, $b=0.0742$, $\eta=0.5638$; MSE is 0.00000062.

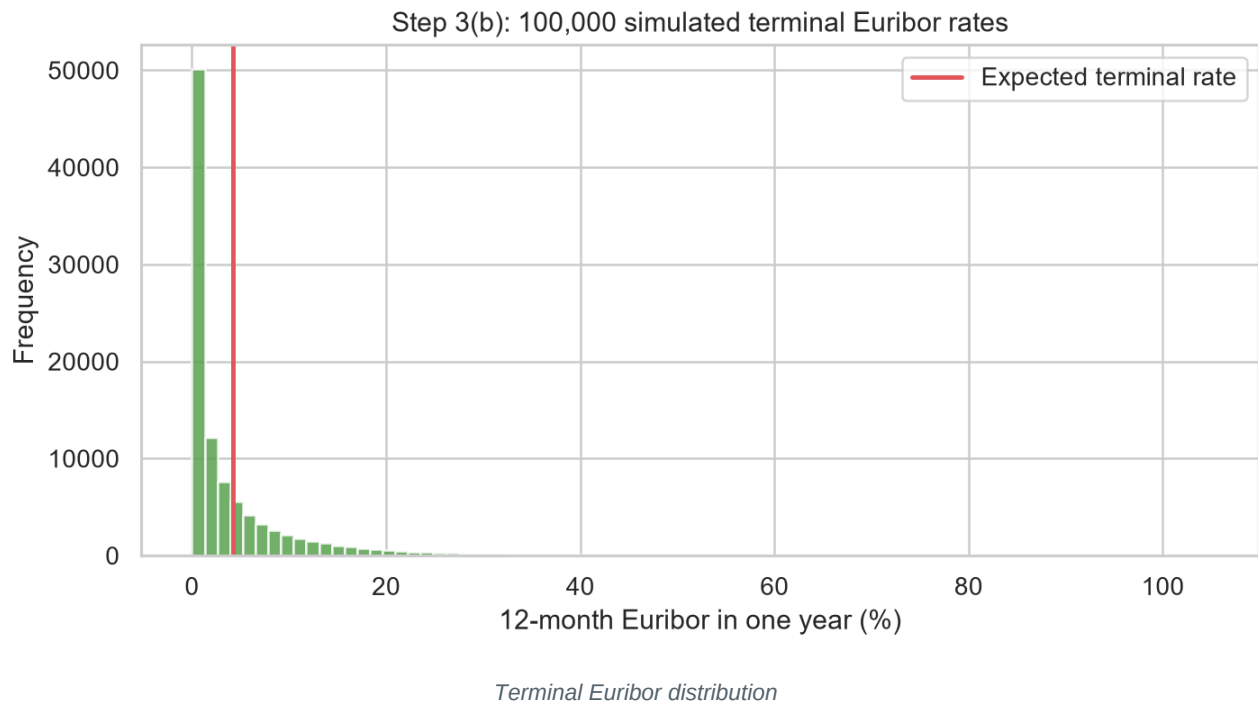


Euribor curve and CIR fit

3(b) One-year rate scenarios

Daily CIR simulation turns the calibrated rate dynamics into a one-year distribution, supporting an expected-rate estimate and an explicit measure of rate uncertainty.

Following the required simulation size (WorldQuant University 4), 100,000 daily scenarios give an expected terminal 12-month Euribor of 4.2289%; the 95% range is 0.0000% to 25.0856%. Higher expected rates increase discounting of future positive cash flows and generally lower present values, all else equal.



Works Cited

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